

# Horn telescope stand assembly

## Teaching guide

### MATERIALS NEEDED

- four 3' long 2 × 4's
- two 2' long 2 × 4's
- two 2' long 2 × 4's with 45° cut at one end for support brace
- 2 ½" construction screws – for assembling the base stand
- 3 ½ " long ¼-20 bolts & wing nuts – for connecting cradle to the base
- Washers

### ASSEMBLING THE BASE STAND

The bottom pieces of the base consists of 2 × 4's notched together, enabling the base pieces to fit together like a log cabin for easy assembly/disassembly.

1. Cut saddle notches in the 2 × 4's as illustrated below. Each notch is 1 ½ " (the width of a 2×4) by 1 ¾ " (half the height of a 2×4).



2. Use the 2 ½ " screws to attach two of the 3' long 2×4's to the other two 3' long 2×4's at 90° at a distance of 10 ½ " from one end.
3. Attach the 2' long 2×4's with the 45° end cuts to the two 3' long 2×4 pieces, as illustrated below.



# FINAL CONSTRUCTION OF STAND WITH HORN ATTACHED:

The final structure is illustrated below:

